Fake News Detection and Evaluation with Confusion Matrix

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1. Abstract

1)Spreading fake news has become a fast and major problem nowadays.
2)It influences the public opinion which causes internal chaos in the society.
3)This project focuses on developing a machine learning model that can automatically classify news articles as **fake** or **real**.
4)Data samples taken from provided two csv files (fake.csv,true.csv). We used textual data (title and content) as the primary features.
5)We evaluated the model's effectiveness using standard metrics to analyse true vs false positives and negatives.

2. Introduction

1)The datasets consist of news articles from Reuters.com, reputable mainstream news provider for true articles and various platforms identified by politifact and wikipedia which are known for spreading false information for fake articles.
2)I have received training on the basics of python that are needed for machine learning fundamentals and basics of machine learning,LLM's and communication skills.

3. Project Objective

1)The main objective of this project is to develop a machine learning model to classify the news articles into fake or true automatically.
2)Another objective is to understand the complete pipeline of machine learning including stages like data preprocessing,text transformation,model training,evaluation,optimization and also to compare the efficiency of machine learning algorithms like logistic regression and decision tree classifier.

4. Methodology

1) **Data Collection :**
The data samples which are used for this project are from fake.csv and true.csv collected from the websites like wikipedia and reuters.com . The articles primarily cover topics related to politics and world news. Each record includes information like title,text,subject,date.

2) **Data Loading :**
We imported the pandas library to load the fake.csv and true.csv into dataframes fake_df and true_df. We created a class column which assigns 0 to fake articles (fake_df) and 1 to true articles (true_df) . We verified these steps by printing the output of each step/cell.

3) **Merging and Shuffling of dataframes :**
Now we merged both the dataframes into a single dataframe named df using concat function and shuffled them with the sample(frac=1) and also used random_state = 0 to reproduce the same shuffled rows every time. We verified it by printing out the output.

4) **Cleaning Data :**
We created a new dataframe called df_clean by removing the first column which contains headings like title,subject,date from df using drop ([‘title’,’date’,’subject’]) and also we resetted the index for the new data Frame df_clean using reset_index(drop=true) and again verified by printing the output.

5) **Text Preprocessing :**
We defined a new function to remove all the URL’s,non-alphanumeric characters,converts the string to lower case, replaces multiple spaces with a single space. We named the function as wordopt(text) and applied to our data frame clean_df text column using df_clean[‘text’].apply(wordopt).

6) **Feature and target splitting :**
Now we defined feature x as ‘text’ column and target y as ‘class’ column and separated our test data from training data with a size of 0.25(25%) using Train_test_split into x_train,x_test,y_train,y_test.

7) Transforming into vectors :

Importing tfidfvectorizer from sklearn.feature_extraction.text to fit and transform the x_train into xv_train using fit_transform(x_train) and only transform the x_test into xv_test using transform(x_test) so that our model does not memorize the test data and only learns the pattern from the training data.

8) Train and Evaluate a logistic regression model :

Importing logistic regression model from sklearn.linear_model to train and we trained the model using fit(xv_train,y_train) and also made predictions on xv_test. To print the classification report we imported classification_report from sklearn.metrics and printed it using classification_report (y_test,predict_lr).

9) Train and Evaluate a Decision Tree Model :

Import decisiontreeclassifier from sklearn.tree and again trained the model just like the above one and imported accuracy_score from sklearn.metrics and made predictions and printed the accuracy score of this model.

10) Hyperparameter tuning using GridSearchCV :

As logistic regression used default parameters we should search for better parameters to improve the efficiency of our model. So we imported gridsearchcv from sklearn.model_selection and performed a grid search operation on logistic regression model. And then atlast we printed out the best parameters using grid.(best_params_).

Workflow of the Project :

Data Collection→Data Labeling→Merge and Shuffle Dataset→Data Cleaning→Text Preprocessing→Feature and Target Selection→Train-Test Split→TF-IDF Vectorization→Logistic Regression Model→Decision Tree Model→Model Evaluation→GridSearchCV Hyperparameter Tuning→Final Results and Analysis

5.Data Analysis and Results

Data Set Summary :

Data Set	Class Label
Fake.csv	**1**
True.csv	**0**

Training and Testing Split :

Data Set Portion	Percentage
Training Data	**75%**
Testing Data	**25%**

Feature Extraction Analysis :

The textual news articles were transformed into vectors using TFIDF technique (Term Frequency Inverse Document Frequency).

Advantages of TF-IDF:

1) Identifies important words within articles.
2) Reduces the influence of common words.
3) Produces numerical features suitable for machine learning algorithms.

Machine Learning Model Analysis :

Two machine learning models were developed and evaluated:

Logistic Regression :

Logistic Regression was trained using the TF-IDF transformed training dataset.

Performance was evaluated using:

1) Precision
2) Recall
3) F1-Score
4) Classification Report

Decision Tree Classifier :

A Decision Tree Classifier was trained on the same training dataset and evaluated using testing data.

Performance was measured using:

1) Accuracy Score

Model	Evaluation Metric	Performance
Logistic Regression	Precision	0.99
	Recall	0.99
	F1 Score	0.99
Decision Tree Classifier	Accuracy Score	~0.99

Hyperparameter Optimization Results :

GridSearchCV was used to identify the optimal parameters for the Logistic Regression model.

Parameters Tested:

1) C = [0.1, 1, 10]
2) Solver = ['liblinear', 'lbfgs']

Best Parameters Obtained:

Parameters	Best Value
C	10
Solver	liblinear

Conclusion :

The experimental results show that machine learning techniques can efficiently identify fake and real news articles. TF-IDF successfully transformed textual information into numerical features suitable for model training. Both Logistic Regression and Decision Tree models achieved high classification performance on the testing dataset.Hyperparameter optimization further improved the performance of the Logistic Regression model.

Confusion Matrix :

The confusion matrix was generated using the predictions of the model to visualize the number of correct and incorrect classifications.

Logistic Regression confusion matrix :

```
from sklearn.metrics import confusion_matrix

cm_lr = confusion_matrix(y_test, prediction_lr)
print(cm_lr)
```

```
[[5360   69]
 [  68 5728]]
```

	Predicted Real news	Predicted Fake News
True Real news	**5360**	**69**
True Fake News	**68**	**5728**

Interpretation:

1) 5360 real news articles correctly predicted as real.
2) 69 real news articles incorrectly predicted as fake.
3) 68 fake news articles incorrectly predicted as real.
4) 5728 fake news articles correctly predicted as fake.

Decision Tree Classifier confusion matrix :

```
cm_dt = confusion_matrix(y_test,prediction_dt)
print(cm_dt)
```

```
[[5398   31]
 [  22 5774]]
```

	Predicted Real news	Predicted Fake News
True Real news	**5398**	**31**
True Fake News	**22**	**5774**

Interpretation:

1) 5398 real news articles correctly predicted as real.
2) 31 real news articles incorrectly predicted as fake.
3) 22 fake news articles incorrectly predicted as real.
4) 5774 fake news articles correctly predicted as fake.